

Problem

Find the rms value of the periodic current

$$i(t) = 8 + 30 \cos 2t - 20 \sin 2t + 15 \cos 4t - 10 \sin 4t \text{ A}$$

Step-by-step solution

Step 1 of 1

Given periodic current

$$i(t) = 8 + 30 \cos(2t) - 20 \sin(2t) + 15 \cos(4t) - 10 \sin(4t) \text{ A}$$

The rms value of the periodic current is,

$$I_{rms} = \sqrt{a_0^2 + \frac{1}{2} \sum_{n=1}^{\infty} (a_n^2 + b_n^2)}$$

$$I_{rms} = \sqrt{(8)^2 + \frac{1}{2} [(30)^2 + (-20)^2 + (15)^2 + (-10)^2]}$$

$$I_{rms} = \sqrt{64 + \frac{1}{2} [1625]}$$

$$I_{rms} = \sqrt{64 + 812.5}$$

$$I_{rms} = \sqrt{876.5}$$

$$\therefore \boxed{I_{rms} = 29.61 \text{ A}}$$